W7 – Intersection of Planes Unit 6
MCV4U
Jensen

SOLUTIONS

Determine the solution(s) to each system of planes. If there are no consistent solutions, give evidence and or an explanation of why.

a) $\pi_1: 3x-2y+z=4$
$\pi_2: 6x-4y+3z=7$

a) normals are not parallel; ∴ intersect in a line.

$2\times①$ $6x-4y+2z=8$
$6x-4y+3z=7$ −
$-z=1$
$z=-1$

Let $x=t$

$3t-2y+(-1)=4$
$3t-1-4=2y$
$3t-5=2y$
$\frac{3}{2}t-\frac{5}{2}=y$

$l: \begin{cases} x=t \\ y=\frac{3}{2}t-\frac{5}{2} \\ z=-1 \end{cases}$

$[x,y,z]=[0,-2.5,-1]+t[1,1.5,0]$

b) $\pi_1: 2x-8y-6z-2=0$
$\pi_2: -x+4y+3z-5=0$

b) normal ARE parallel; check if coincident or distinct.

① $2x-8y-6z-2=0$
$2\times②$ $-2x+8y+6z-10=0$ +
$-12=0$

∴ NO solutions
Parallel and distinct.

c) $\pi_1: x-2y+2.5z-1=0$
$\pi_2: 3x-6y+7.5z-3=0$

d) $\pi_1: x+y+2z=5$
$\pi_2: 4x-3y+z=-8$
$\pi_3: -5x-2y+3z=7$

e) $\pi_1: x+y+z=-3$
$\pi_2: 2x+2y-3z=4$
$\pi_3: 3x+3y-2z=1$

f) $\pi_1: -x+y+3z=2$
$\pi_2: 2x-2y-6z=-4$
$\pi_3: -3x+3y+9z=6$

g) $\pi_1: x-4y-13z=4$
$\pi_2: x-2y-3z=2$
$\pi_3: -3x+5y+4z=2$

h) $\pi_1: 3x-2y+5z=1$
$\pi_2: 5x+y-3z=-4$
$\pi_3: x-5y+13z=6$

i) $\pi_1: -x+y+3z=2$
$\pi_2: -x+y+3z=4$
$\pi_3: x-3y+5z=6$

j) $\pi_1: x+4y+3z=5$
$\pi_2: x+3y+2z+4=0$
$\pi_3: x+y-z=-1$

c) normals are parallel; check if distinct or coincident

$3\times$ ① $3x-6y+7.5z-3=0$

② $3x-6y+7.5z-3=0$ −

$0=0$

∴ parallel and coincident

∴ infinite solutions.

d) no normals are parallel

① $x+y+2z=5$

$2\times$ ② $8x-6y+2z=-16$ −

$-7x+7y=21$

$3\times$ ① $3x+3y+6z=15$

$2\times$ ③ $-10x-4y+6z=14$ −

$13x+7y=1$

$-7x+7y=21$

$13x+7y=1$ −

$-20x=20$

$x=-1$

$13x+7y=1$

$13(-1)+7y=1$

$7y=14$

$y=2$

$x+y+2z=5$

$-1+2+2z=5$

$2z=4$

$z=2$

The intersect at the point $(-1,2,2)$

e) $2\times$ ① $2x+2y+2z=-6$

② $2x+2y-3z=4$ −

$5z=-10$

$z=-2$

$3\times$ ① $3x+3y+3z=-9$

③ $3x+3y-2z=1$ −

$5z=-10$

$z=-2$

let $x=t$

① $x+y+z=-3$

$t+y+(-2)=-3$

$y=-1-t$

$$l:\begin{cases} x=t \\ y=-1-t \\ z=-2 \end{cases}$$

$[x,y,z]=[0,-1,-2]+t[1,-1,0]$

f) $-2\pi_1=\pi_2$ Three coincident planes.

$3\pi_1=\pi_3$ Infinite solutions.

g) No parallel normals

① $x-4y-13z=4$

② $x-2y-3z=2$ −

$-2y-10z=2$

$3\times$ ① $3x-12y-39z=12$

③ $-3x+5y+4z=2$ +

$-7y-35z=14$

$7\times(①-②)$ $-14y-70z=14$

$2\times(3\times①+③)$ $-14y-70z=28$ −

$0=-14$

∴ No solutions. It is an inconsistent system.

h) $5\times$① $15x-10y+25z=5$

$3\times$② $15x+3y-9z=-12$ −

④ $-13y+34z=17$

① $3x-2y+5z=1$

$3\times$③ $3x-15y+39z=18$ −

⑤ $13y-34z=-17$

④ $-13y+34z=17$

⑤ $13y-34z=-17$

$0=0$

∴ they intersect in a line. Let $y=t$

$-13t+34z=17$

$34z=17+13t$

$z=\frac{17}{34}+\frac{13}{34}t$

③ $x-5y+13z=6$

$x-5t+13\left(\frac{17}{34}+\frac{13}{34}t\right)=6$

$x-5t+\frac{221}{34}+\frac{169}{34}t=6$

$x=-\frac{1}{2}+\frac{1}{34}t$

$l:\begin{cases} x=-\frac{1}{2}+\frac{1}{34}t \\ y=t \\ z=\frac{17}{34}+\frac{13}{34}t \end{cases}$

$[x,y,z]=\left[-\frac{1}{2},0,\frac{17}{34}\right]+t\left[\frac{1}{34},1,\frac{13}{34}\right]$

i) $\vec{n}_1=\vec{n}_2$ but $\pi_1\neq k\pi_2$; ∴ parallel and distinct.

∴ an inconsistent system.

j) on back.

Answers:

a) $[x,y,z]=\left[\frac{5}{3},0,-1\right]+t\left[\frac{2}{3},1,0\right]$
b) No solutions; parallel and distinct
c) Infinite solutions, parallel and coincident
d) (-1,2,2)
e) $[x,y,z]=[-1,0,-2]+s[-1,1,0]$
f) infinite solutions; three coincident planes
g) inconsistent system; the planes intersect in pairs
h) $[x,y,z]=\left[-\frac{7}{13},-\frac{17}{13},0\right]+t\left[\frac{1}{13},\frac{34}{13},1\right]$
i) inconsistent system; two parallel and distinct planes intersected by a third plane
j) [redacted] (-52,30,-21)

j) ① $x+4y+3z=5$
② $x+3y+2z=-4$ −

④ $y+z=9$

⑥ $x+4y+3z=5$
③ $x+y-z=-1$ −

⑤ $3y+4z=6$

$3\times$④ $3y+3z=27$
⑤ $3y+4z=6$ −

$-z=21$

$z=-21$

& intersect at a point

④ $y+(-21)=9$

$y=30$

① $x+4(30)+3(-21)=5$

$x=-52$

They intersect at $(-52, 30, -21)$

$10=6+3t$
$0=-2-t$
$z=-3-2(-2)$

$(3,0,0)$
$(0,0,1)$